

Sandra Sudevan
192424422

SET 1

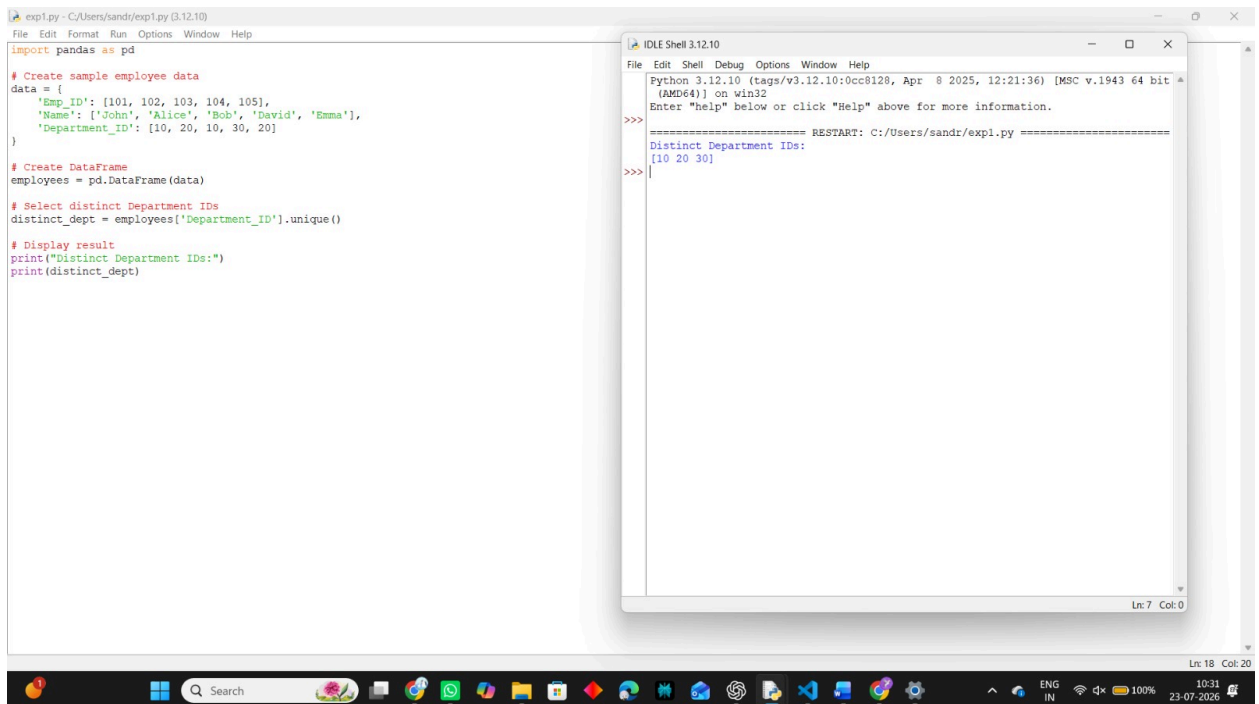
1. Write a Pandas program to select the distinct department id from employees file.

Aim:

To write a Pandas program to select the distinct department IDs from the employees file.

Procedure:

1. Import the Pandas library.
2. Create or read the employees dataset.
3. Use the `unique()` function to find distinct department IDs.
4. Display the result.



The screenshot displays a Python IDE with two windows. The left window, titled 'expl.py - C:/Users/sandri/exp1.py (3.12.10)', contains the following code:

```
import pandas as pd

# Create sample employee data
data = {
    'Emp_ID': [101, 102, 103, 104, 105],
    'Name': ['John', 'Alice', 'Bob', 'David', 'Emma'],
    'Department_ID': [10, 20, 10, 30, 20]
}

# Create DataFrame
employees = pd.DataFrame(data)

# Select distinct Department IDs
distinct_dept = employees['Department_ID'].unique()

# Display result
print("Distinct Department IDs:")
print(distinct_dept)
```

The right window, titled 'IDLE Shell 3.12.10', shows the output of the program:

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/sandri/exp1.py =====
Distinct Department IDs:
[10 20 30]
```

The Windows taskbar at the bottom shows the date and time as 23-07-2026, 10:31.

Result:

The distinct department IDs were displayed successfully.

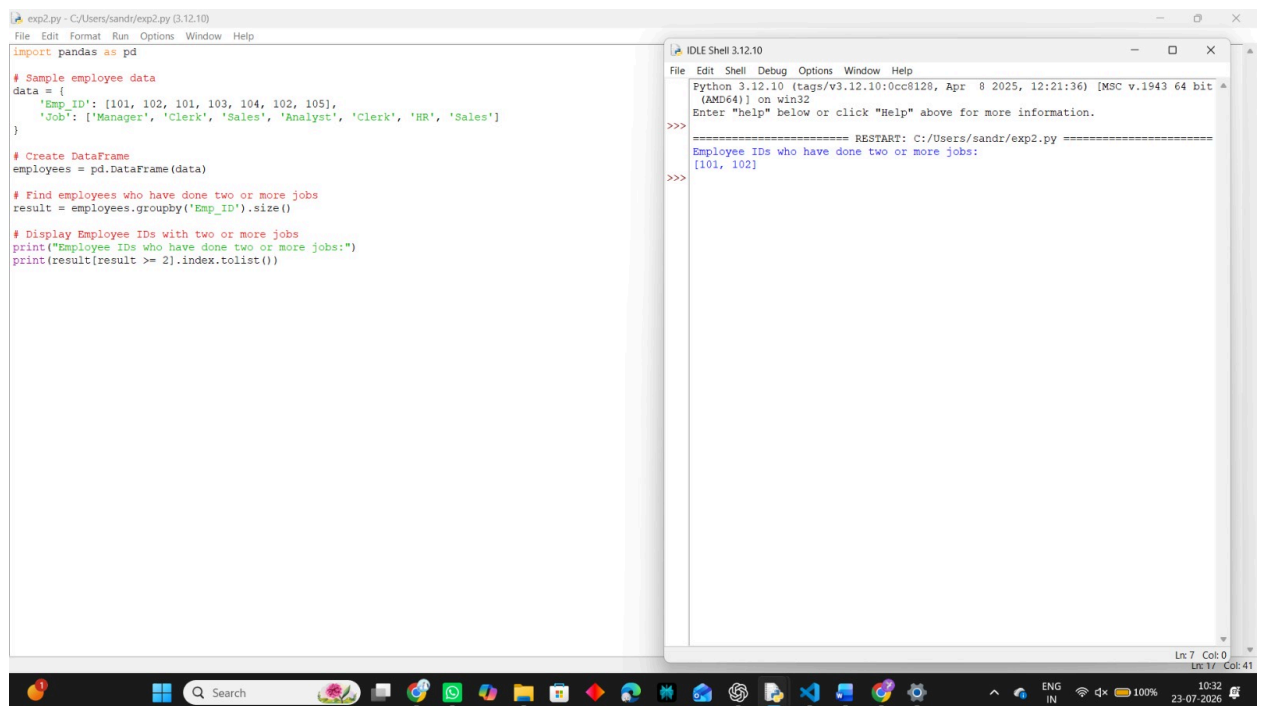
2. Write a Pandas program to display the IDs of employees who have done two or more

Aim:

To write a Pandas program to display the IDs of employees who have done two or more jobs.

Procedure:

1. Import the Pandas library.
2. Create or read the employee dataset.
3. Group the records using `groupby()`.
4. Count the jobs done by each employee.
5. Display employee IDs with two or more jobs.



The screenshot shows a Python IDE with two windows. The left window, titled 'exp2.py - C:/Users/sandr/exp2.py (3.12.10)', contains the following code:

```
import pandas as pd

# Sample employee data
data = {
    'Emp_ID': [101, 102, 101, 103, 104, 102, 105],
    'Job': ['Manager', 'Clerk', 'Sales', 'Analyst', 'Clerk', 'HR', 'Sales']
}

# Create DataFrame
employees = pd.DataFrame(data)

# Find employees who have done two or more jobs
result = employees.groupby('Emp_ID').size()

# Display Employee IDs with two or more jobs
print("Employee IDs who have done two or more jobs:")
print(result[result >= 2].index.tolist())
```

The right window, titled 'IDLE Shell 3.12.10', shows the output of the program:

```
>>>
===== RESTART: C:/Users/sandr/exp2.py =====
Employee IDs who have done two or more jobs:
[101, 102]
>>>
```

The Windows taskbar at the bottom shows the date as 23-07-2026 and the time as 10:32.

Result:

The employee IDs who have done two or more jobs were displayed successfully.